BuyerUnion | Visa audit and concise PRD

CalHacks 13.0 | Research checked October 23, 2026 | Planning, not a tested implementation

1. Recommendation and challenge fit

Proceed with a buyer-led laptop marketplace: merchant agents compete through complete offers, then an independent evidence layer helps the buyer choose. The strongest demonstration connects agent reasoning to enforced merchant limits, real source citations, and explicit purchase approval.

Confirmed: four teammates; no existing code or inventory; custom buyer UI; new and refurbished laptops/accessories; U.S./USD, San Francisco delivery area, buyer-selected deadline. Seek real authorized inventory first; use clearly labeled sample inventory with real web evidence if reliable/full data is unavailable. Try Visa sandbox; an explicitly labeled simulation is approved as fallback. Skip Fetch AI if ASI:One is mandatory. Entire packet stays below ten pages.

Visa rubric audit

Potential below is a design assessment, not a measured judging score. The weights come from your supplied rubric.

Criterion	Strongest fit	What could undermine it
Innovation - 25%	Buyer intent triggers independent, inventory-aware merchant offers and an evidence review.	Reverse quotations already exist. A scripted council with ordinary catalog results adds little novelty.
User Experience - 25%	Broad and precise requests converge on a few clear options.	Long debates, slow scraping, excessive questions, and unexplained rankings recreate research work.
Technical Execution - 20%	Scoped tool-using agents, policy checks, asynchronous evidence, recoverable order state.	Agents invent terms; callbacks duplicate actions; a winning offer disappears.
Commerce Impact - 15%	Less buyer comparison effort and merchant quoting work; suitable inventory can win on service.	No merchant willingness, negative contribution after fees, or unfulfilled offers.
Trust & Responsibility - 15%	Private limits, source provenance, exact-model matching, independent evaluation, bounded approval.	Scraped claims presented as facts; merchant evidence cherry-picking; implied real merchant authority.

Critical changes: keep the council bounded and optional to watch; verify mandatory specifications before ranking; treat final sentiment as decision support; reserve and recheck the selected quote after research. A factual mismatch can disqualify an offer. Negative sentiment alone should not silently override the buyer's requirements.

Novelty boundary: Alibaba RFQ, Amazon Business RFQ, and Google Direct Offers establish adjacent quotation, competition, and customized-offer mechanisms. Differentiate through a complete consumer journey and reliable execution, not the presence of agents alone. [2-4]

2. Sponsor targets and the recommended stack

Five primary category candidates; a conditional sixth through a custom Apify Actor. These are meaningful submission targets, not confirmed eligibility or guaranteed permission to combine awards. Track/prize details below come from your supplied sponsor list. Places within one competition are not separate integrations.

Track / listed prize	Useful implementation	Qualification or access gate
Visa - Best use of Reimagining Shopping; $5,000	Approved-offer manifest and sponsor-supported sandbox payment step.	Ask which API/SDK and operation qualify. Public event listing does not establish an API requirement. Simulation must remain labeled.
Apify - Best use of Apify; $2,000 / $1,000 / $800 / $400 / $400	Retrieve real finalist evidence using maintained Actors; retain source and run metadata.	Prove chosen sources work, control spend, and confirm event submission requirements.
Convex - Best use of Convex; $1,308 heading, hardware in description	Sole source of truth for roles, requests, inventory, policies, offers, jobs, and reservations.	Use real transactions, authenticated functions, and live progress; confirm sponsor proof requirements.
MLH - Best Use of Gemini API; swag kits	Direct Gemini API for request clarification, merchant tool use, and source-grounded synthesis.	Actual API use; inspect structured output and validate commercial values in code.
MLH - Best Use of Vultr; Tab5	Host an actual application or agent/connector worker needed by the demo.	Requires an account and running workload. Do not create an idle server just for a logo.
Apify - Best Actor; $400, conditional	Reusable Product Evidence Pack Actor: exact models in, normalized source-linked evidence out.	Confirm whether a new/public Actor, Store publication, or extra form is required. Existing Actor consumption alone does not establish qualification.
Elastic - Agent Builder on Serverless; $1,200 / $1,436 headings, hardware descriptions; optional	Read-only independent evidence verifier over public evidence.	Must use Agent Builder on Serverless, not search alone. Add after the core works; index is a derived research copy, not authoritative stock.

Keep: custom UI + Convex + Gemini API + Apify + approved Visa adapter; Vultr runs a necessary worker if that deployment is selected. Convex Workflow can manage durable steps; add a separate pool only if concurrency needs it. [7-12]

Fetch AI: its official CalHacks brief requires Agentverse registration, Agent Chat Protocol, ASI:One discovery, and the full primary workflow within ASI:One. Your custom-UI preference therefore excludes the Fetch track under the current rules. Its prize amounts also conflict with Devpost; neither discrepancy needs to change this decision. [5]

Spacetime: a credible alternative to Convex, with transactional reducers and private data/views. Do not run two authoritative databases. Its sponsor blog mentions a special prize absent from your supplied list; category, amount, and eligibility remain unconfirmed. [15]

3. Remaining sponsor tracks: coverage without feature bloat

Every remaining supplied sponsor family is screened here. Add only a useful, demonstrated integration and verify its own rules. Prize headings sometimes describe hardware, credits, or mixed packages; do not total them as cash.

Track family	Fit / possible use	Decision
Figma / Figma Agent: design, use-case, creative categories	Design the vague-intent flow, comparison cards, and evidence drawer. Agent categories need actual Figma Agent use.	Low-runtime-cost candidate; confirm category rules.
Factory - Factory Droid SDK	Actual SDK-powered merchant data-import or repository-validation workflow.	Optional. Coding with the CLI alone does not prove SDK use. [13]
Cartesia	Voice intake with editable transcript and written final approval.	Optional after text flow; avoid spoken bidding noise.
MLH - GoDaddy Registry domain	Eligible project domain through the specified route.	Low-runtime-cost candidate; confirm suffix/rules. No registration purchase is authorized by this report.
MLH - Commit Fellowship	Working team prototype, real problem, opt-in and supplemental form.	Worth considering if pursuing the startup; not automatic admission.
Moss	Potential local/private semantic retrieval for catalog or evidence.	Verify event sponsor identity/rules; skip redundant retrieval stores.
Creao AI - AI-First Disruptor / One-Person Company / Team Empowerment	Merchant operator automation has thematic fit.	Required CREAO usage and category definitions unverified; do not duplicate the core workflow.
Microsoft - Built for the Real World / Customer Impact	A real merchant outcome could fit.	Brief does not identify qualifying Microsoft technology; ask rather than invent requirements.
Hyphenate - Autonomous Systems for Finance	Merchant margin governance or post-order reconciliation is adjacent.	Ask for finance brief. Shopping debate alone is insufficient evidence of fit.
Coinbase - Best Use of x402	Agent-paid, metered evidence service could fit.	Defer wallet/payment complexity. Keep distinct from buyer Visa checkout; unnamed second track is unresolved.
CRS - Overall / Financial AI Agent / Identity-Fraud / Unexpected	Merchant identity or finance expansion.	Weak core fit; credit data does not establish product truth. Access and qualification unknown.
Bitdeer AI Model Studio; MLH Snowflake API; MLH Tiger Data	Alternative model serving, research, or data backend.	Alternatives to chosen components, not extra layers storing the same thing.
MLH Solana; Capital One Nessie	Additional settlement or mock banking/budget functions.	No necessary role in the current flow; avoid redundant payment simulations.
Armadin - Best Security Hack	Isolation and injection-resistance could support a security-focused demonstration.	Ask its brief; ordinary safeguards alone do not establish track eligibility.
Meta/Facebook - Bringing People Closer Together with AI	Possible future shared purchasing.	Current commerce flow has weak social-connection fit; do not bolt on a social network.
SpaceXSI; Notability	No necessary role in the laptop marketplace.	Skip. SpaceXSI's supplied brief requires space data and Cursor.

General awards: Overall, Most Creative Project, Most Technically Advanced Hack, and Hacker's Choice need no extra stack. Beginner eligibility is unknown. People-First / Planet-First need substantive impact evidence; refurbished stock alone proves no environmental saving. Hardware has no current fit; Solo does not fit the confirmed four-person team.

Organizer gates: confirm sponsor-category/award-combination limits, additional submission portals, and deadline details. The event header says October 25, 2026, 9:00 AM PDT; body text uses PST inconsistently. The live pitch is three minutes plus two minutes of questions; repository, image, and demo are required. [1]

4. Integration audit: what connects and what remains yours

Flow: custom buyer UI -> authenticated Convex request -> scoped merchant agents -> validated offers -> bounded council -> Apify evidence jobs -> neutral comparison -> selected-offer recheck/reservation -> buyer approval -> Visa sandbox or labeled simulation.

Layer	Reuse	Team-owned integration and guardrail
State and orchestration	Convex transactions, subscriptions, Workflow	Bind buyer/merchant/service identities; enforce tenant access; version requests and offers. External actions are not database transactions.
Merchant agents	Gemini API tool calls and structured outputs	One isolated context per merchant. Tools read own catalog/policies and propose offers. Backend rejects forbidden prices, terms, substitutions, or stock claims. JSON shape does not prove semantic correctness.
Inventory	Authorized merchant export/API where available	Store model, configuration, unit/condition, source, freshness, stock status, price components, delivery and allowed concessions. A public listing grants no permission to negotiate for that seller.
Evidence	Apify asynchronous run -> dataset -> validated callback/poll	Persist run ID, exact product identity, status, query, source date, retrieval date, Actor/build and source URL. Deduplicate; handle blocked, empty, partial, and timed-out results.
Final evaluation	Gemini synthesis; optional Elastic specialist	Separate manufacturer specifications, merchant promises, independent tests, and owner reports. Match revision/configuration/region. A model review cannot prove a refurbished unit's battery health.
Payment	Actual sponsor-granted Visa operation	Approve exact merchant, item/configuration, quantity, total, expiry, and offer revision. Report the operation actually performed; tokenization alone is not a paid order.

Reliability and trust boundaries

- **Council contract:** each merchant gets a bounded claim, moderator challenge, response, and optional revision/abstention. Merchant agents see their own data and permitted claim context, not rivals' cost floors or the buyer ceiling. The moderator validates rather than awarding the loudest argument.
- **Final research:** collect shared neutral evidence for the distinct models in the final 3-5 offers. Show themes, disagreement, freshness, source count, identity caveats, and links. A claimed owner is not a verified buyer. Site-provided verified-purchase labels must be attributed to that site.
- **Untrusted text:** source pages cannot rewrite merchant policy, authorize tools, expose secrets, or approve payment. Check citations actually support the corresponding claim.
- **Async recovery:** persist state before external calls; deduplicate event/run IDs; retry safely and reconcile ambiguous remote success. Convex's base scheduled actions are not automatically retried. Database atomicity cannot roll back external inventory holds or Visa operations. [8-11]
- **Truthful modes:** authorized live merchant; public catalog snapshot with demo merchant authority; synthetic demo inventory; cached real evidence; Visa sandbox; simulated checkout. Display the actual mode. Do not imply real seller participation from a scraped storefront.

Main remaining dependency: an authoritative merchant source with permission to submit quotes and, ideally, reserve stock. Without it, demonstrate agent and local reservation behavior honestly in demo mode.

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Core Details

Purpose: help people buy a suitable laptop/accessory bundle with less comparison work, while reducing merchant quoting effort. Four-person hackathon prototype; custom buyer UI; new/refurbished products; U.S./USD, San Francisco area, buyer-selected deadline. Core accounts: buyer, merchant, and service worker.

Scope: broad or precise intake; editable constraints; three merchant agents as the proposed demo target; complete offers; bounded council; final 3-5 results where enough qualify; independent web evidence; selected-offer revalidation; explicit approval; Visa sandbox attempt with labeled simulation fallback. Real authorized inventory first; approved sample fallback. No claim of live merchant participation without authority.

Non-goals for this build: unrestricted categories, live-money purchases, consumer credit, escrow, guaranteed savings, open-ended negotiation, and cross-merchant all-or-nothing checkout. One merchant may supply a multi-item bundle. Commercial rollout and revenue terms remain open.

Why

Buyers currently connect requirements, specifications, price, delivery, protection, and reviews themselves. Merchants repeat quotation work. The product makes offers comparable and research inspectable. Innovation depends on useful tailored terms and reliable execution. Merchant arguments are advocacy; the independent evidence layer and buyer decision remain separate.

Flow

1. **Buyer intake:** accept a broad goal or exact specification. Ask only material clarifications; show editable must-haves/preferences and private budget. Confirm deadline and delivery area.

2. **Merchant setup:** connect/upload permitted stock; confirm data freshness, allowed bundles, price limits, fulfillment, warranty, and returns. Merchants can pause agents or require approval for exceptions.

3. **Offer formation:** authenticated merchant agents read only their own inventory/policy, call tools, and propose itemized offers or abstain. Server checks requirements and authority before accepting a bid. First bids remain private.

4. **Council:** one bounded challenge/response round over disclosed claims and requirements; optional merchant-approved improvements. Show concise progress and expandable reasons rather than forcing users to read a transcript.

5. **Final research:** Apify gathers independent evidence for qualifying finalists. Present exact-model matches, recurring reports, disagreement, citations, and missing evidence. Repeated copies count as one source family. No artificial consensus percentage.

6. **Decision:** factual mismatches return to validation; sentiment informs the buyer. If fewer than three offers qualify, show the true count. Never relax mandatory requirements silently.

7. **Purchase handoff:** recheck stock, expiry, delivery and final charges; obtain a hold if supported. Buyer approves the current offer revision. Invoke the actual Visa sandbox capability or labeled simulation and show an accurate receipt/state. A changed quote requires approval again.

Integration contract: request/version -> merchant-scoped offer/policy version -> council events -> research job/run/evidence -> finalized quote/reservation -> approval -> payment/order events. Every external job carries correlation and duplicate-handling identifiers. No scraper callback can approve a purchase or alter a merchant policy.

Impact

Expected benefits are hypotheses: lower comparison effort, fewer unsuitable purchases, less merchant quoting work, and profitable incremental orders. No conversion, savings, or environmental uplift is claimed without measurements.

Acceptance check	Proposed demonstration target
Broad and precise needs	Both produce confirmed structured requests and understandable results.
Merchant reasoning is real	Three distinct policy/inventory contexts call tools; one permitted revision and one abstention/rejection are observable.
Commercial limits	All prepared below-floor, wrong-specification, and unauthorized-term cases are rejected.
Evidence integrity	Displayed claims link to matching sources; wrong-model evidence is excluded; sparse evidence is labeled.
Workflow reliability	Duplicate callbacks cause no duplicate transition; expired quotes require reapproval; the last sample unit cannot be reserved twice.
Purchase truthfulness	User approval binds the final quote; sandbox/simulation and remote-hold limitations remain visible.
Value measurement	Record task time, feasible-offer count, checkout changes, merchant contribution assumptions, and source coverage against a simple catalog baseline.

These are test targets, not completed checks. Measure cold-run and cached-run latency separately; set a user-facing wait budget after the first real scrape. An older cached result retains its retrieval timestamp and must be labeled.

Implementation Plan

Milestone	Deliverable / exit condition	Proposed owner
M0: access and data gate	Confirm sponsor rules, keys/quotas, a test product/source, merchant authority and reservation capability. Decide live versus fallback modes explicitly.	All four
M1: thin end-to-end path	Custom intake -> one merchant offer -> stored state -> comparison -> approval -> demo ending. Backend enforces identity and price/spec rules.	A + B + D
M2: agents and evidence	Three tool-using merchant contexts, bounded council, asynchronous Apify evidence, exact-model matching and source drawer.	B + C
M3: execution and recovery	Chosen-offer revalidation, native/remote hold distinction, Visa sandbox or simulation, replay/timeout/expiry tests.	D + B
M4: submission-ready	Repeated live run, clearly labeled cached fallback, short video, image/repository, concise pitch, actual integration proof. Add optional tracks only after this path works.	All four

Four workstreams: A owns buyer/merchant UI and interaction design; B owns merchant policy/tools/council; C owns Apify, product identity and evidence; D owns Convex, access control, payment adapter and deployment. Assign people based on skills, which remain unknown. Milestones are dependencies, not a promised calendar schedule.

Resources: Convex deployment/auth setup; direct Gemini API access; Apify account, working Actors and capped budget; sponsor-provided Visa sandbox credentials/test data if available; Vultr only for a genuine deployed workload; authorized inventory or labeled fallback. Do not place API secrets in browsers, model prompts, or public repositories.

Reuse instead of rebuilding: existing Actors, Convex realtime/Workflow and transactions, Gemini tool-calling/schema facilities, sponsor payment examples, and a small inventory adapter. The team still builds commercial authority checks, product identity, council rules, source interpretation, and approval semantics.

7. Decisions, remaining questions, and sponsor asks

Decided with you

- Ten-page maximum; four teammates; no existing code/inventory.
- Laptop/accessory focus, new and refurbished; custom buyer UI.
- U.S./USD, San Francisco delivery area, deadline chosen by the buyer.
- Real reliable/full inventory preferred; sample fallback plus real web evidence permitted.
- Try Visa sandbox; clearly labeled simulation if unavailable.
- Skip Fetch AI when its mandatory ASI:One flow conflicts with the chosen UI.

Proposed directions needing team confirmation

- Three merchant contexts for the demo; one bounded council round; private budgets and commercial floors; no mandatory discounting.
- Final research is advisory on sentiment; disqualify only substantiated factual/mandatory-constraint failures and return to validation.
- Convex as the sole source of truth; Gemini as core reasoner; Apify as extraction; optional Elastic specialist only after core stability.
- One merchant per purchased bundle; no real-money transactions in this prototype. Actual seller responsibility, fees, and support policies remain commercial decisions.

Questions still open

1. **Merchant access:** which merchant can supply authorized inventory and permit agent quotations? Are minimum prices, condition/unit evidence, and reservation actions available? What measurable condition standard is acceptable for refurbished units?

2. **Source access and cost:** which exact models and public sources work through tested Actors, what may be stored/displayed, and what budget/quotas does each sponsor grant?

3. **Implementation resources:** what are the four teammates' strongest languages/frameworks, and who owns each workstream? Which authentication and hosting route can the team set up reliably?

4. **Interaction decisions:** approve the bounded council and source-advisory policy above? Should the buyer see price/service tradeoffs or a default suggested offer with alternatives?

5. **Commercial model:** who handles cancellations/returns/support; what fees are permitted; what is merchant participation worth after costs? None is validated yet.

Ask sponsors / organizers before committing integrations

- **Visa:** which operation, SDK/API, test credentials and approved sandbox path are available today; is integration mandatory; how should an unavailable-access fallback be presented?
- **Apify:** which uses qualify, what credits and source access are offered, and what exactly qualifies for Best Actor? Is publication or a supplemental submission required?
- **Convex / Gemini API / Vultr:** confirm credits, plan/runtime limits, actual required usage and submission evidence. For Elastic, confirm Agent Builder on Serverless access and supported tools.
- **Organizers:** category-entry and award-combination limits, supplemental forms, check-in/deadline timezone. For optional Figma/Factory/Cartesia categories, obtain their actual briefs first. Do not infer qualification from a sponsor logo.

Execution checklist: idea audited against all five Visa criteria; all supplied sponsor families screened; integration boundaries checked against current docs; PRD includes Core Details, Flow, Impact, Why, and Implementation Plan; unanswered decisions surfaced. No product code, sponsor integration, source scraper, or merchant connection was executed in this audit.

TLDR: build the custom-UI marketplace around enforced merchant agents, independent finalist evidence, and bounded purchase approval. Target five primary sponsor categories, with a custom Apify Actor and Elastic as conditional additions. Documentation supports the architecture; data, credentials, eligibility and real-world impact remain unverified. Resolve access first, then prove one complete laptop purchase journey.

8. Sources and verification limits

Source priority: your supplied sponsor/prize text and scoring rubric; current official event/sponsor briefs for eligibility; Context7 and primary technical docs for capabilities. Statements about potential fit and proposed behavior are recommendations. Product documentation is not proof of event eligibility or successful integration.

1. CalHacks 13.0: event listing, requirements and prizes and FAQ: implementation during hacking period; ideation allowed beforehand. Rules page exposed the Code of Conduct, not a verified award-stacking policy.

2. Alibaba RFQ: buyer requirements and supplier quotations.

3. Amazon Business RFQ: qualifying bulk requests, offers and checkout.

4. Google Direct Offers: merchant guardrails, tailored bundles and UCP checkout descriptions.

5. Fetch AI official CalHacks hackpack: ASI:One, registration, ACP, deliverables; prize discrepancy recorded.

6. Visa Intelligent Commerce and developer quick start: authorized instructions/tokenization and sandbox/production distinctions.

7. Convex transactions and actions: local state versus external calls.

8. Convex scheduled functions and Workflow: delivery/retry distinctions and persisted orchestration.

9. Convex HTTP actions and authentication in functions: validate inputs and enforce identity/authorization.

10. Apify Actor runs, running Actors, and webhooks: asynchronous extraction and completion.

11. Apify publishing: platform publication instructions do not establish Best Actor event rules.

12. Gemini structured outputs and tools: typed responses and application-executed calls; semantic validation still required.

13. Factory Droid SDK: actual programmatic harness, tools and permissions; distinguish SDK use from CLI-assisted development.

14. Elastic custom agents and Agent Builder APIs: custom application access; named track requires Serverless.

15. Spacetime at CalHacks, reducers, and procedures: alternative architecture; unlisted-prize uncertainty.

Verification: the attachment and public sources were read; current technical documentation was queried. No credentials, account eligibility, runtime behavior, live inventory, source extraction, price savings, or checkout integration was tested. Sponsor credits and quotas are not guaranteed. Optional-track screening reflects the supplied brief, not undisclosed rules. Prize headings with hardware/credit descriptions are not treated as cash. Final PDF page count and layout are checked separately from product feasibility.